Information Retrieval and Extraction

PageRank and LexRank

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Query and Result Set Matching

Query Representation

- Problem Definition
  - Given a news article, A_i published on T day
  - Generate all the *relevant* tweets of A_i from all the Tw published on T
- Challenges
  - Huge *vocabulary gap* between A_i and T
  - Direct matching of keywords is not enough!
  - Usage of *tweet* specific content
- Iterative *relevance* matching
- Representation of A_i
  - Need to distinguish A_i from all A_j published on T
  - Identify the top 10 *tf-idf* scored *Nouns* and *Adjectives*
    - *IDF* is based on all A_j published on T, *distinguishing* factors
    - *Nouns* and *Adjectives* worked better empirically
    - Could consider any *word* or *keyphrases* or *entity* or *verb included*

Similarity Calculation

TF-IDF based

- TF IDF assumes a bag of words (BoW) representation
- A_i is represented by the set of words present, w_i
- Generate a vocabulary and article matrix for T day which comprises of
  - Words, W as rows and articles, A as columns
  - TF score of the word (*can be specific, nouns and adjectives*) is the frequency of the word in that A_i
  - IDF (actually, the log of the IDF) gives more importance to words which appear rarely across articles
- Find top most 10 tf-idf scored words of every article
  - If *threshold* number of these words occur in T_j
  - Consider T_j

Similarity Calculation

TF-IDF based

- Consider A_i and T_j is similar if $S(i,j)$ is greater than threshold,

$$S(i,j) = \frac{\sum_k tf(A_i^k) tf(T_j^k) idf_k^2}{\sqrt{\sum_o (tf(A_i^o) idf_o)^2} \sqrt{\sum_p (tf(T_j^p) idf_p)^2}} \quad (1)$$

where, k, o and p are the unique nouns and adjectives of (A_i and T_j), A_i and T_j, respectively

For MMR based

- Can we use the same score to calculate similarity between *query* and *document* ?
  - *query* can be only unique nouns and adjectives (no, notion of TF-IDF)
- To calculate similarity among *documents* ?

Similarity Calculation

Real-life Application based

- Challenge of *huge vocabulary* gap still remains
  - Iterative *similarity* calculation across tweets
  - Usage of *tweet* specific ranked content and context

Similarity Calculation

News Article	Initial T	Extended H	Extended T
Clinton offers new plan for tuition-free college	embracing sanders' higher-ed call, clinton expands tuition-free college plan - i'm not biting the bait	*#bernieforHillary*, *#tuitionfree*	a political olive branch? do u believe she will give debt-free college??

Similarity Calculation

News Article	Initial T	Extended H	Extended T
Britain's planning for Iraq War wholly inadequate:Chilcot Report	scathing chilcot report slams tony blair over botched iraq war	*#chilcot*, *#chilcotreport*	remember that blair said history will be my judge well, it's just happened. and the judgement is utterly damning.

Similarity Calculation

News Article	Initial T	Extended H	Extended T
Oscar Pistorius gets 6 years in prison for murdering girlfriend	wow. only six years. oscar pistorius sentenced to six years jail	*#oscarpistorius* *#reevasteenkamp* *#masipaisusless*	someone killed a chicken in front of a church. got sentenced to 25 years in prison he gets sentenced to 6yrs for manslaughter

Similarity Calculation

News Article	Initial T	Extended H	Extended T
Obama says 8,400 troops will remain in Afghanistan	president obama says troop levels in afghanistan will stay the same, setting aside the drawdown he once promised	*#obamabetrayus*	obama-clinton are both *reckless* in promoting wars for profit. why don't they send their own kids to fight in afghanistan? hypocrites.

Similarity Calculation

Frequency based

- Frequency based similarity between T_i and T_j

$$S_{ij} = \frac{\sum_{I \in w_i \cap w_j} F(I)}{\sum_{U \in w_i \cup w_j} F(U)} \quad (2)$$

- Are all words equally important?
  - What about hashtags?

Examples

- Tweets
  - *embracing sanders' higher-ed call, clinton expands tuition free college plan* #hillary #tuitionfees #uselection #Isupporthillary
  - *clinton supports free education, tuition free college!!!* #Isupporthillary
- $S_{ij} = ?$

Similarity Calculation

Hashtag Based

- Similarity between T_i and T_j

$$S_{ij} = \frac{\sum_{I \in \mathcal{H}_i \cap \mathcal{H}_j} F(I)}{\sum_{U \in \mathcal{H}_i \cup \mathcal{H}_j} F(U)} \quad (3)$$

- Are all hashtags equally important?

Examples

- Tweets
  - *embracing sanders' higher-ed call, clinton expands tuition free college plan* #hillary #tuitionfees #uselection #Isupporthillary
  - *a political olive branch? do u believe she will give debt-free college??* #neverhillary #tuitionfees #uselection
- $S_{ij} = ?$

Similarity Calculation

Hashtag Based

- Identify *important* and *specific* hashtags
  - A hashtag is *specific*, $\mathcal{H}^s$ to the event if it has not occurred before and represents only a specific sub topic, ex #tuitionfees
  - $\mathcal{H}^g$ could be the most frequent hashtags that has previously occurred in the previously last N months of the occurrence before the event or subevent or occur consistently across subevents, ex #neverhillary, #uselection, #hillary, #Isupporthillary
- $\mathcal{H}^s = \mathcal{H}$- $\mathcal{H}^g$
  - A *specific* hashtag also co occurs with other *specific* hashtags frequently
- Similarity between T_i and T_j

$$S_{ij} = \frac{\sum_{I \in \mathcal{H}_i^s \cap \mathcal{H}_j^s} F(I)}{\sum_{U \in \mathcal{H}_i^s \cup \mathcal{H}_j^s} F(U)} \quad (4)$$

Similarity Calculation

Embedding based

- Word2Vec can capture contextual similarity
  - Learns the interrelationship between a pair of words based on a huge corpus
  - Visualize a model being trained on all available English documents dataset, ex: google news
- Intuition
  - Co-occuring words are similar
  - Words that co-occur with same/similar words are similar

Similarity Calculation

Summary of Insights

- Pre processing
  - Removal of irrelevant words, such as, stop words, noise
  - Consider only specific and relevant information, such as, key phrases, hashtags
- Determination of threshold on the basis of S_{ij}
  - Might be combined, must be normalized
- Requires understanding of application specific data

Google Ranking Algorithm

PageRank Centrality

- A *webpage* is important
  - if its *referred* in multiple web-pages, so ranked *higher*
  - more **important** websites are likely to receive **more links** from other websites
- Purpose of "measuring" its relative importance wrt other web-pages
- How to generate a ranking algorithm?
  - Propose a *link analysis* algorithm
  - Visualize all *web-pages* as nodes
  - There is a *directed* edge from W_i to W_j if W_j is referred in W_i
- PageRank works by counting the *number* and *quality* of links to a page
  - Provides a rough estimate how important this website is

Google Ranking Algorithm

Degree based Centrality

- One who has most friends is most important
  - One who has most web-pages linked is most important
- Summation of the number of connections : *Influencer*
  - *Undirected* : Facebook, LinkedIN
  - *Directed* : Twitter, Instagram
- Degree centrality

$$C_i = \frac{deg_i}{N-1} \tag{5}$$

- There is a high variance in degree centrality scores across nodes
  - Power law degree distribution!
- Are all connections equally important?

Google Ranking Algorithm

PageRank Centrality

- My neigbors are important, so am *important*
- PageRank assigns nodes a score based
  - their connections
  - their connections' connections

$$C_i = \sum_{j \in Nei(T_i)} \frac{C_j}{deg_j} \tag{6}$$

- Transformation into Page Rank Centrality

$$P_i = \frac{1-d}{N} + d \sum_{j \in Nei(T_i)} \frac{P_j}{deg_j} \tag{7}$$

Google Ranking Algorithm

PageRank Centrality

- Each node has a score
  - If every node "sends" its score to its neighbors be equal to its original score

$$P_i^{t+1} = \frac{1-d}{N} + d \sum_{j \in N} A_{ij} \frac{P_j^t}{deg_j} \tag{8}$$

- Important nodes are those
  - connected to important nodes
  - Repeat till *convergence*
- Damping Factor
  - Represent the likelihood that a person randomly clicking on links will arrive at any particular page
  - an imaginary surfer randomly clicking on links
    - person will continue following links is a *damping factor*, d
    - instead jump to any random page is $(1-d)$
    - generally, d is kept at 0.85

Applications : Tweet Summarization

Introduction

- Majority of the user population prefer social media platforms than traditional news media platforms
- The huge popularity of online social networks has affected the news consumption behavior of users
- A large fraction of users consume news, share their opinions and discuss related to news on online social networks, like Twitter
- Studying news events and user opinions towards the news events have several applications
  - Understand the popularity of the news event
  - Understand user preferences and stance towards particular news
  - Determine the evolution of the news story

Applications : Tweet Summarization

Introduction

- Challenges
  - Enormous volume, presence of noise and redundant information
  - Mining of the important perspectives immensely difficult
- Automated tweet summarization of the users' reactions and opinions
  - Understanding the sensitivity/importance
  - Insights about the diverse opinions
  - Major aspects that attracts the users attention
- Objective : Identify the subset of tweets such that each tweet
  - *Relevance* : relevant to the event or timeline
  - *Diversity* : capture the diverse opinions
  - *Coverage* : represent all the user opinions

Applications : Tweet Summarization

Graph based Representation

- Let, $G_1 = (V_1, E_1)$ denote a graph of tweets with respect to an event
  - V_1 comprises of all the tweets
  - E_1 represents whether a pair of tweets is similar or not
- Generate a graph where T_i and T_j have an edge if $S(i,j)$ is greater than threshold,

$$S(i,j) = \frac{\sum_k tf(T_i^k) tf(T_j^k) idf_k^2}{\sqrt{\sum_o (tf(T_i^o) idf_o)^2} \sqrt{\sum_p (tf(T_j^p) idf_p)^2}} \quad (9)$$

where, k, o and p are the unique words of (T_i and T_j), T_i and T_j, respectively

LexRank

Introduction

- Importance of a tweet depends on
  - number of tweets its connected
  - importance of the corresponding tweets
- Based on Page Rank centrality

$$L_i = \frac{1-d}{N} + (d) \sum_{j \in Nei(T_i)} \frac{S_{i,j}}{\sum_{k \in Nei(T_j)} S_{j,k}} \quad (10)$$

- Select representative tweets as those tweets which has the highest LexRank Score, L_i
- Iterative integration of similar nodes affect coverage adversely
  - Many tweets of same subtopic increases LexRank Score
  - Impact the representativeness of the tweets that do not belong to this topic

Thank you!